SOFTWARE REQUIREMENTS SPECIFICATION

Money Manager Application

A Professional Python Desktop Finance Application

Built on 3-Tier Architecture + SOLID Principles

Python 3.10+ | SQLite | PySide6 | v1.0

Software Architecture Team · Version 1.0 · March 2024

Project Scope

Desktop Application

Built using Python for Windows, Mac, and Linux platforms

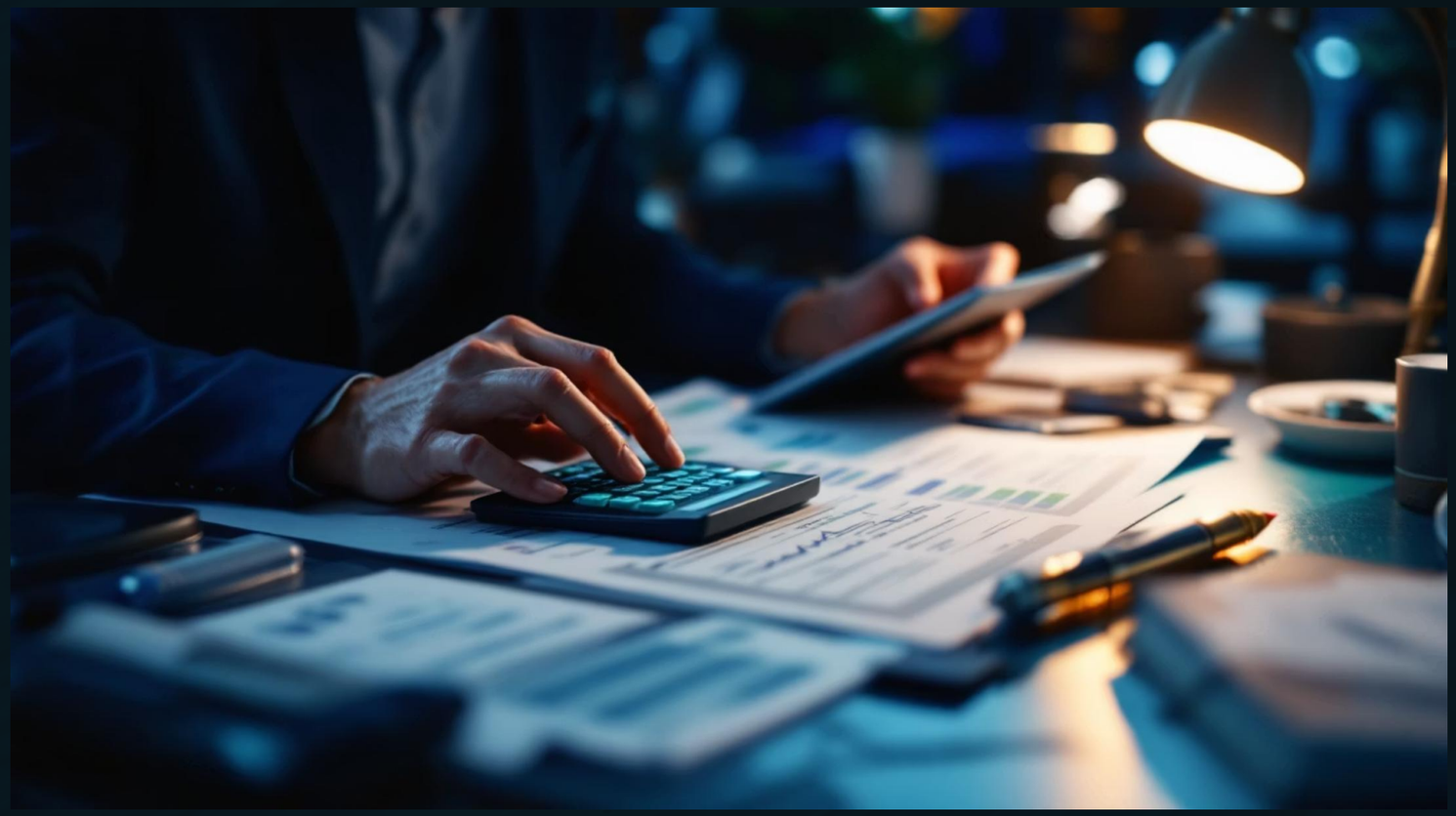
Offline-First

Full functionality without internet connection for privacy and reliability

Data Management

Secure local storage with backup capabilities

Project Overview



Project Purpose

A personal finance management desktop application designed to help users track their financial health with precision and ease.

Core Capabilities

- Track income and expenses in real-time
- Organize and manage monthly budgets
- Generate comprehensive financial reports
- Visualize spending patterns with charts

TECH STACK

Technology Stack

Layer	Technology	Purpose
GUI	PySide6	*Cross-platform desktop UI*
Services	Python 3.10+	*Business logic & validation*
Database	SQLite 3	*Embedded relational database*
Build / Dist	PyInstaller	*Self-contained executable build*

INTRODUCTION

Project Name	Money Manager App
Language	Python 3.10+
Architecture	3-Tier + SOLID Principles
Database	SQLite (local-first)
GUI Framework	PySide6
Future Plans	Android · Web · Cloud Sync

AGENDA

What We'll Cover

01 Introduction & Objectives

02 System Architecture

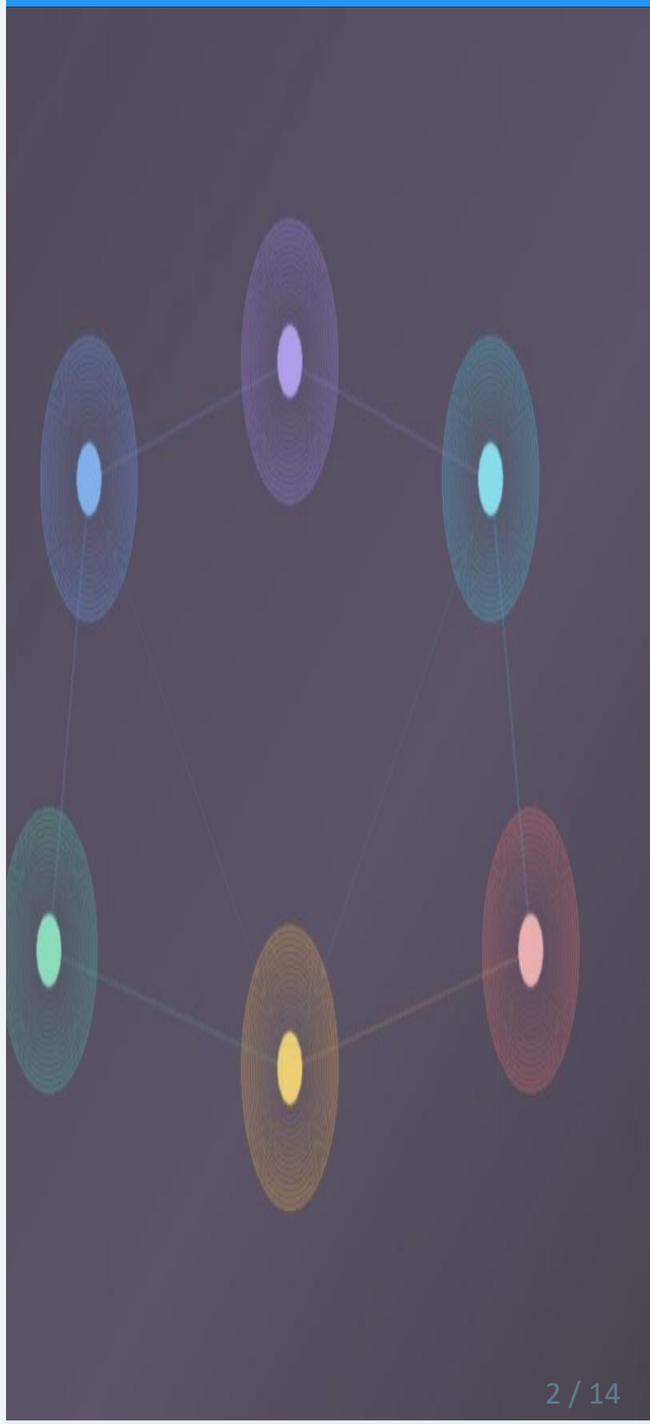
03 SOLID Principles

04 Core Features

05 Database Design

06 Requirements

07 Future Roadmap



OBJECTIVES

Project Objectives

Track Finances

Record expenses with full categorization

Visual Reports

Dashboard with charts and spending trends

Budget Control

Per-category monthly limits with alerts

Goals

Personal reminders and savings goal tracking

Data Integrity

Reliable SQLite storage with ACID compliance

Future-Ready

Clean architecture for Android & Web expansion

ARCHITECTURE

PRESENTATION LAYER

«GUI Layer»

PySide6

▼ *calls / delegates*

BUSINESS LOGIC LAYER

«Service Layer»

TransactionService · BudgetService · UserService

▼ *calls / delegates*

DATA LAYER

«Repository Layer»

user_repo · TransactionRepo · BudgetRepo

SOLID Design Principles

S

Single Responsibility

Each class has one job only

TransactionService
BudgetRepo

O

Open / Closed

Extend features via new classes

not by modifying existing ones

L

Liskov Substitution

SQLiteRepo can swap to APIRepo

without breaking services

I

Interface Segregation

Small focused interfaces only

IBudgetRepo
ITransactionRepo

D

Dependency Inversion

Services depend on abstractions

not concrete implementations

Database Design

Relational database schema with user-centric data organization ensuring data integrity and efficient queries.

1

Users Table

User profiles storing credentials, preferences, and personal information

2

Transactions Table

Financial operations including amounts, categories, dates, and descriptions

3

Budgets Table

Monthly spending limits by category with tracking periods

4

Notes Table

Additional transaction details, comments, and documentation

5

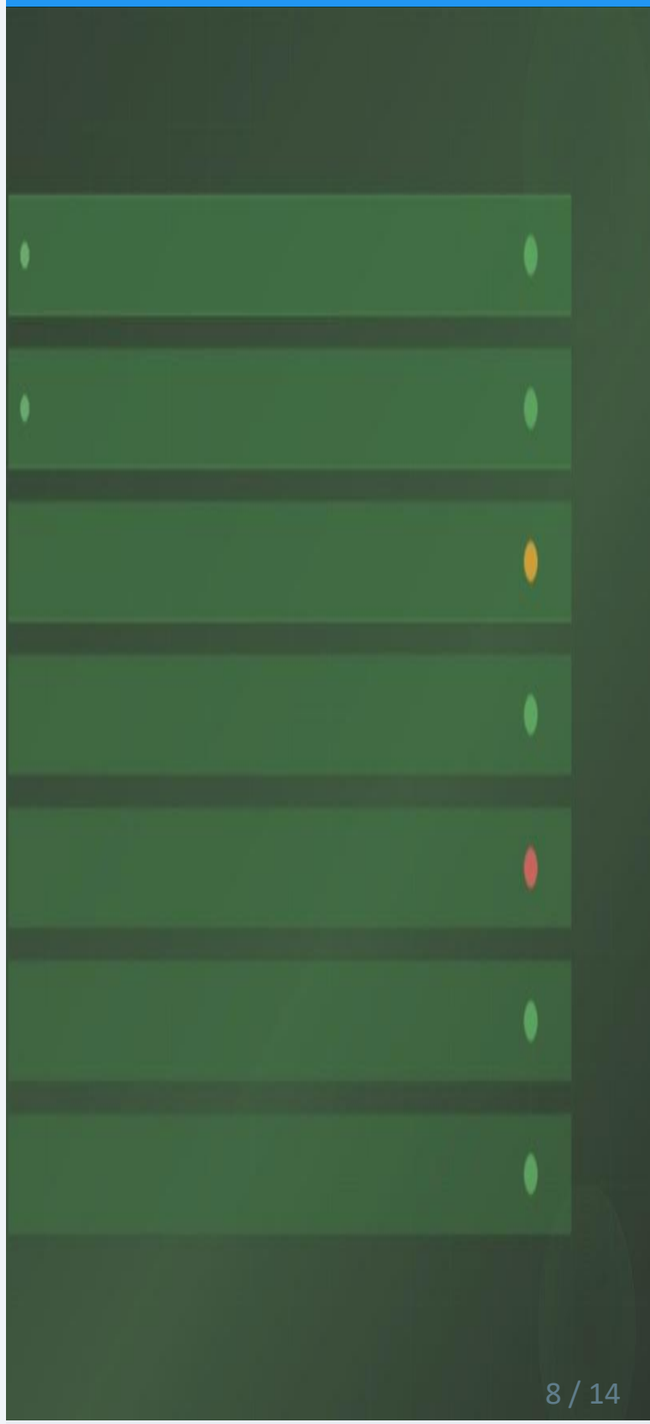
Goals Table

Savings targets with amounts, deadlines, and progress tracking

DATABASE DESIGN

Database Schema

users	*id, username, password , Permissions*
transactions	*id, amount, category, Month, Year*
budgets	*id, amount, Month, Year , total_amount*
Goals	*id, target_amount, saving_amount*



Users

PK User_ID

Username

Password

Permissions

Goals

PK Goal_ID

Name

Target_amount

Saved_amount

Budget

PK Budget_ID

Name

Amount

Month

Year

total_amount

Transactions

PK Transaction_ID

Name

Amount

Category

Month

Year

NFR

Non-Functional Requirements

Performance

< 3s launch · < 200ms UI

Reliability

Atomic writes · no data loss

Usability

Core tasks in < 5 min, no docs

Security

bcrypt hashing · no plaintext

Portability

Windows · macOS · Ubuntu 20.04+

Scalability

SQLite → REST API with 0 changes

Maintainability

> 80% unit test coverage

Data Integrity

FK constraints · cascade deletes

USE CASES

UC-01

Add Transaction

Actor → GUI → Service → Repository → DB
Validates input, persists, refreshes dashboard

UC-02

Monitor Budget

System alerts at 80% and 100% of limit
Real-time spending vs budget comparison

UC-03

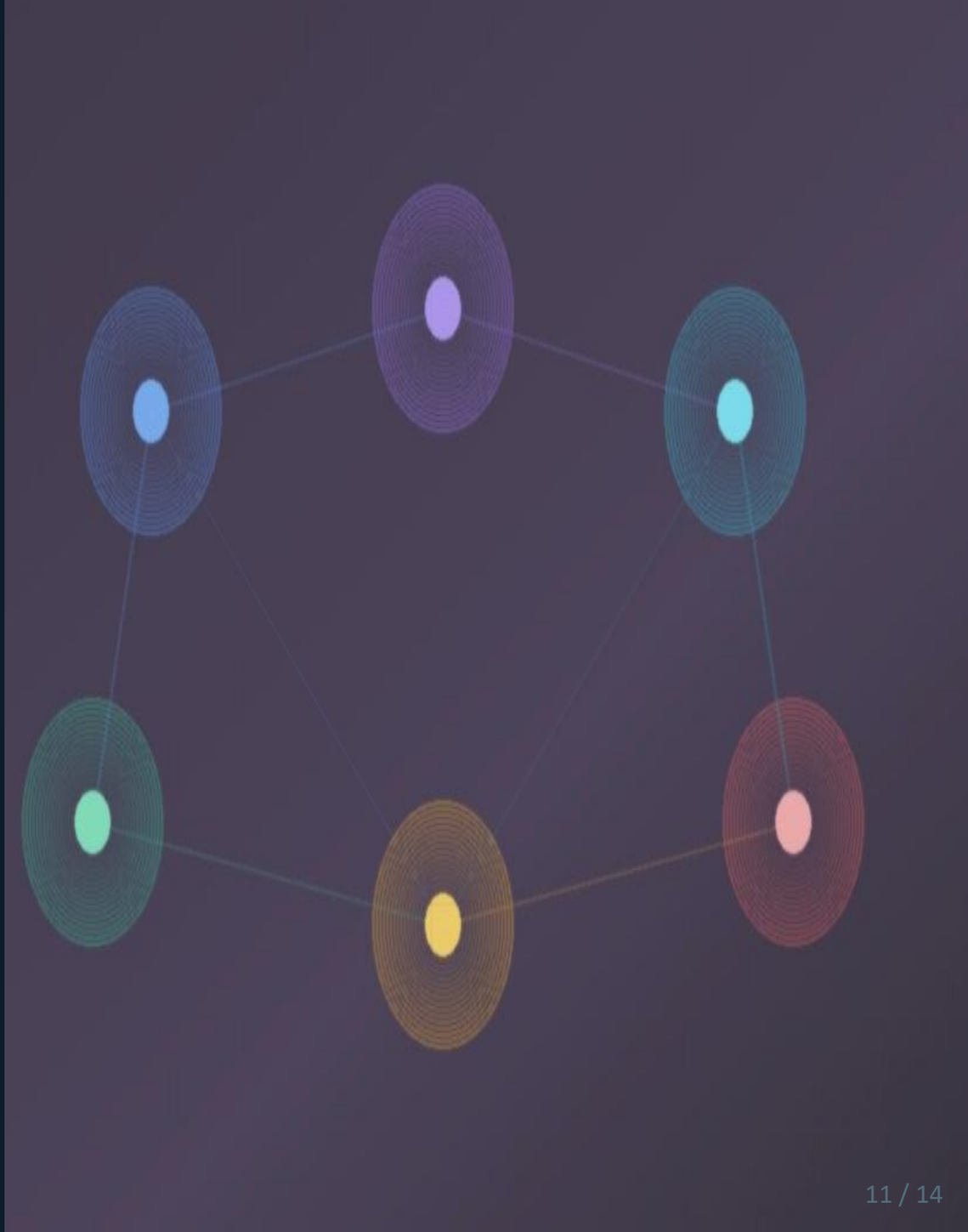
View Dashboard

Aggregates transactions, renders charts
KPI cards with monthly trends

UC-04

Set Savings Goal

Define target amount + deadline
System tracks progress and sends reminders



Key Features

Saving & Expense Tracking

Log transactions with categories, dates, and descriptions. Track cash and digital payments seamlessly.

Monthly Budget Management

Set spending limits by category. Get alerts when approaching or exceeding budget thresholds.

Interactive Dashboard

Real-time overview of financial health with summary cards and quick access to key metrics.

Export Reports

Generate detailed financial reports in PDF and Excel formats for record keeping and analysis.

Multi-Currency Support

Manage finances in different currencies with automatic conversion rates and currency-specific tracking.

Financial Goals

Set savings targets and track progress toward specific financial objectives with milestone tracking.

Users & Stakeholders

1

End User

Individual managing personal finances, tracking daily expenses, and planning budgets

2

Developer

Python engineer building and maintaining application codebase and features

3

Quality Assurance

Tester ensuring functionality meets requirements and identifying bugs before release

4

Product Owner

Project manager defining requirements, prioritizing features, and approving deliverables

System Architecture

A three-tier architecture ensuring separation of concerns and maintainable code structure.

Presentation Layer

User interface built with modern GUI framework. Handles user input, displays data, and manages screen navigation.

Business Logic Layer

Core application logic processing transactions, validating rules, calculating budgets, and generating reports.

Data Layer

Database management system storing all financial data securely with efficient queries and backup mechanisms.

REQUIREMENTS

Functional Requirements

FR-01 · Transactions

- Add / edit / delete transactions
- Filter by date, category, type
- Auto-categorization by keyword

FR-02 · Budgets

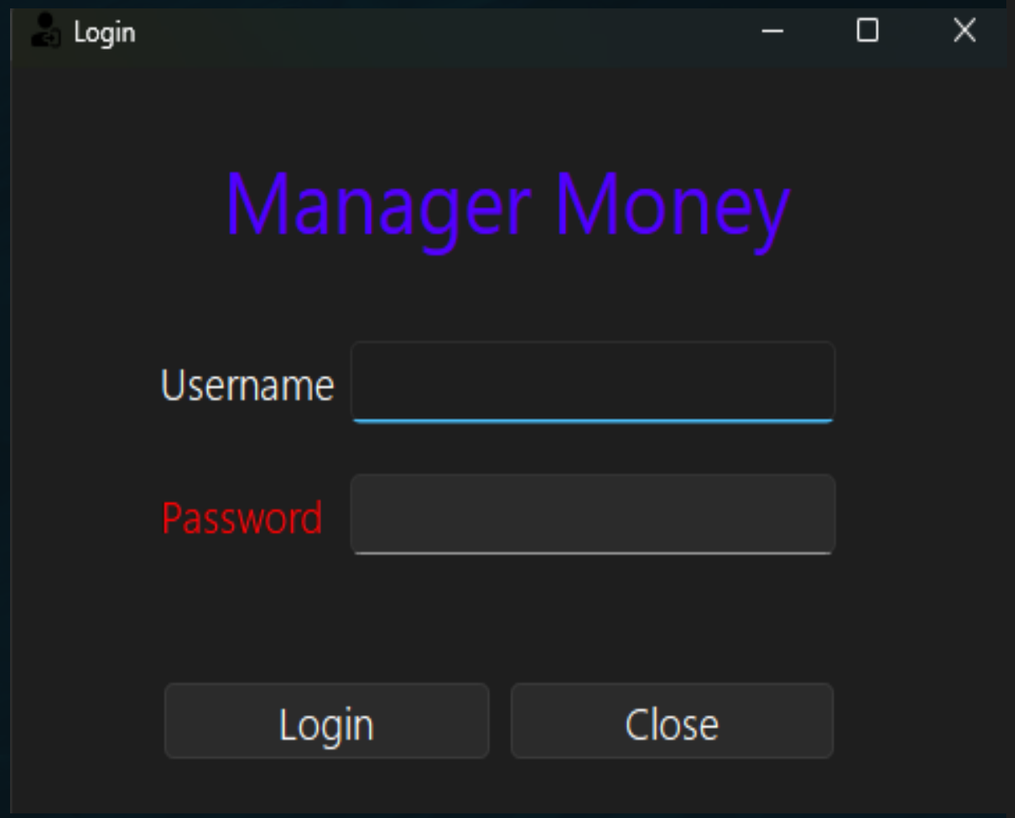
- Per-category monthly budgets
- 80% and 100% over-limit alerts
- Real-time spending progress

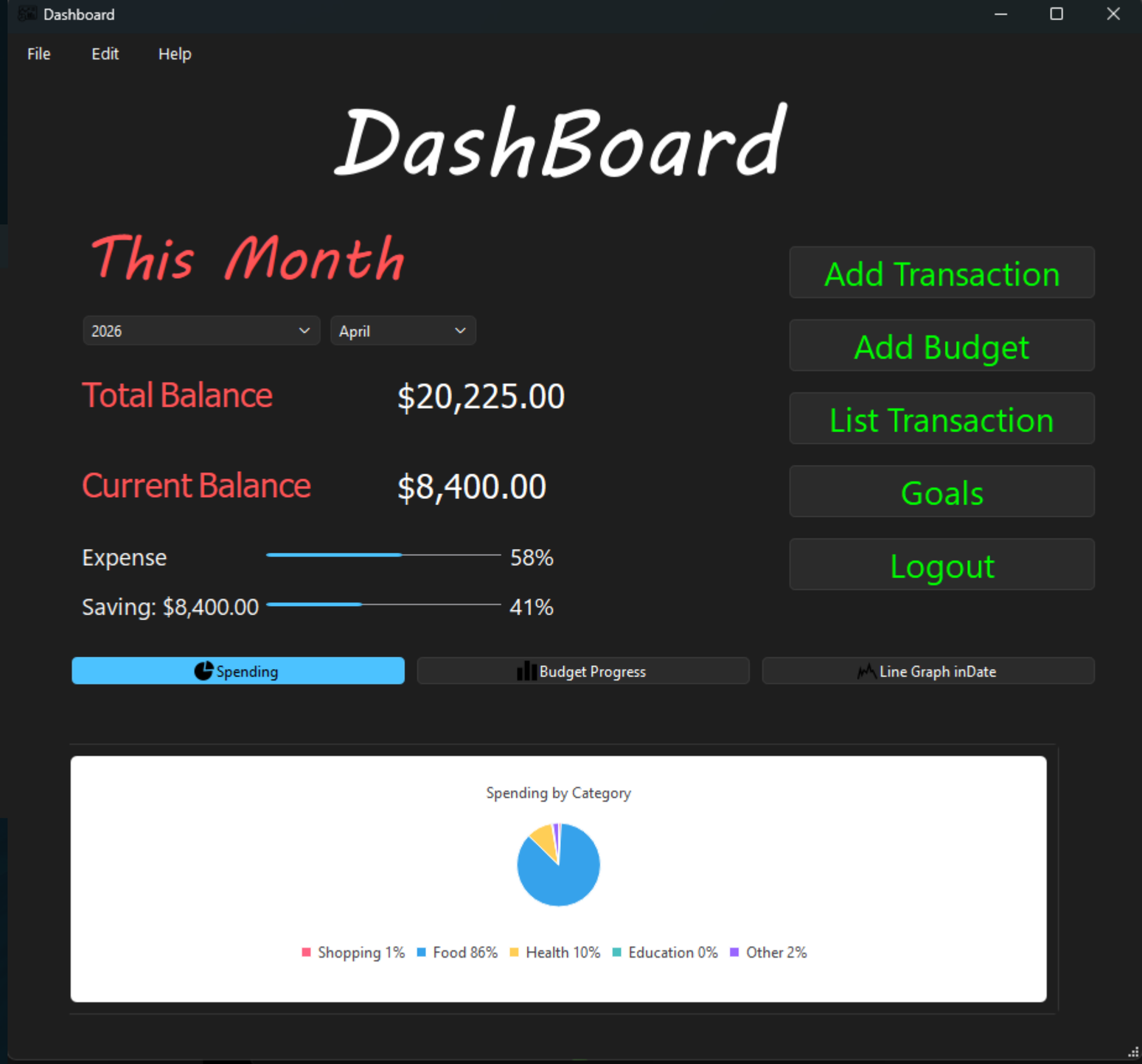
FR-03 · Dashboard

- Monthly KPI summary cards
- Pie & bar chart visualizations
- PDF / Excel report export

FR-04 · Goals

- Tagged notes with reminders
- Savings goals with % progress
- Deadline tracking & alerts


Login
Manager Money
Username
Password
Login
Close


Dashboard
File
Edit
Help
DashBoard
This Month
2026
April
Total Balance
$20,225.00
Current Balance
$8,400.00
Expense
58%
Saving: $8,400.00
41%
Add Transaction
Add Budget
List Transaction
Goals
Logout
Spending
Budget Progress
Line Graph inDate
Spending by Category
Shopping 1%
Food 86%
Health 10%
Education 0%
Other 2%

Transaction

New Transaction

Amount 0000.00

Catagory Food

Month 04

Year 2026

Available Balance 8400.0

Save Close

Add Budget

New Budget

Year 2026

Month Budget 04

Amount 0000.00

Save Close

Savings Goals

My Savings Goals

+ New Goal

lap x

$50.00 of $200.00

25%

Click to view details

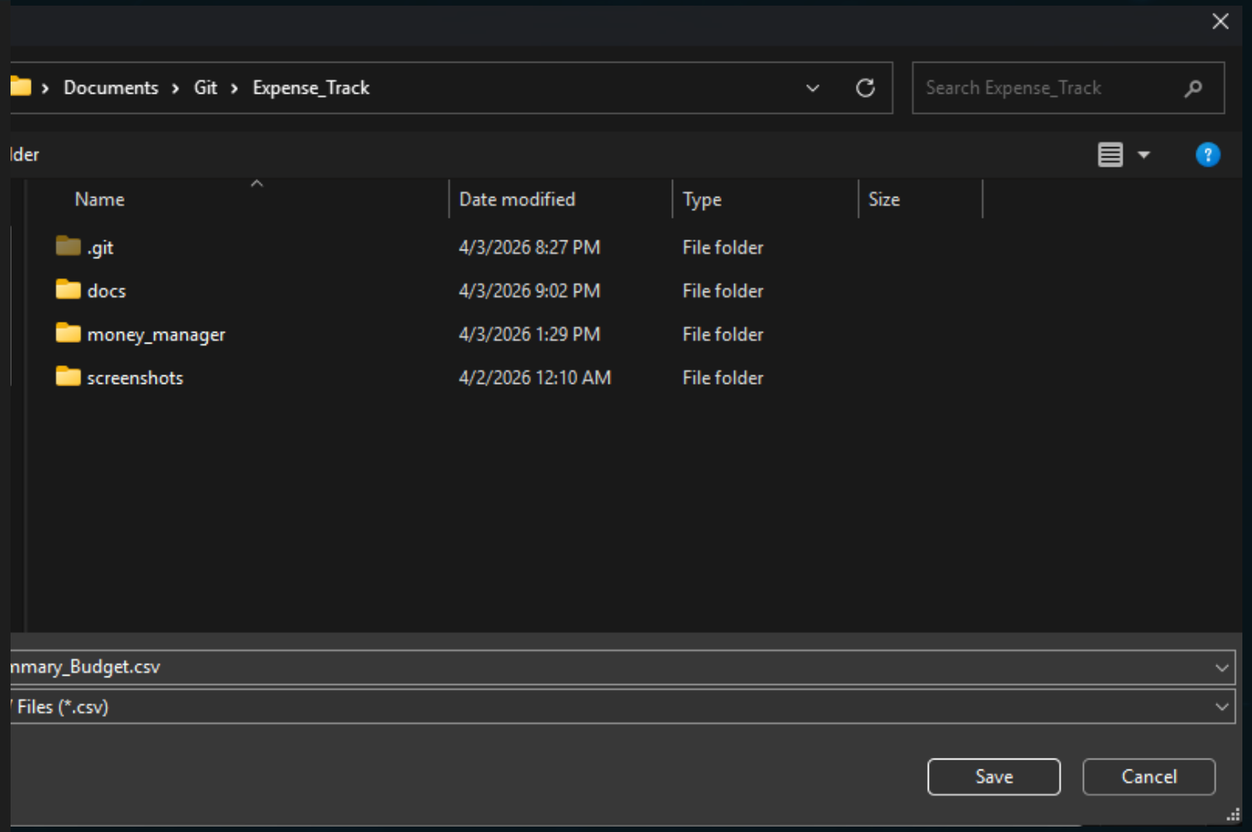
Close

List Transactions

List All Transaction

	ansaction I	Amount	Category	Month	Year
1	7	400.0	Shopping	4	2026
2	9	400.0	Food	4	2026
3	11	4444.0	Health	4	2026
4	12	344.0	Food	4	2026
5	13	400.0	Food	4	2026
6	15	134.0	Food	4	2026
7	16	122.0	Food	4	2026
8	17	1000.0	Food	4	2026
9	18	100.0	Food	4	2026
10	19	200.0	Food	4	2026
11	20	200.0	Education	4	2026

Export Close



Future Enhancements

Mobile App

iOS and Android applications for on-the-go expense tracking

Web Version

Browser-based access with responsive design for any device

Cloud Sync

Automatic data synchronization across all devices and platforms

AI Analysis

Machine learning to predict spending patterns and optimize budgets

ROADMAP

Future Roadmap

Current

- ✓ Python Desktop App
- ✓ SQLite Local Storage
- ✓ PyQt5 / PySide6 GUI

Q3 2026

- ✓ Android App (Kivy)
- ✓ Cloud Sync (Firebase)
- ✓ Multi-user Support

Q1 2027

- ✓ Web App (FastAPI + React)
- ✓ PDF / Excel Reports
- ✓ AI Spending Insights

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